

NB IMCQ 4

Spring 2026

Embryology, Histology & Neuroscience

1. A 12-year-old girl is brought to the physician by her mother because of a 2-month history of a painless, enlarging mass in her neck. Physical examination shows a smooth round mass the size of a golf ball located just anterior to the sternocleidomastoid. A diagnosis of internal branchial sinus is suspected. If this mass has an internal opening, it is most likely to open into which of the following areas?

- A. Posterior 1/3rd of tongue
- B. Tonsillar area
- C. Foramen cecum
- D. Pharyngeal recess
- E. Vestibule of larynx

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SOM.MKII.BPM2.1.NB.1.ANAT.0209: Describe the development of the branchial sinuses, fistulas, and cysts

A 4-month-old boy is admitted to the hospital for surgical repair of a facial developmental defect as shown in the photograph. This condition is most likely to develop from a failure to fuse which of the following embryological structures?

- A. Maxillary and medial nasal process
- B. Opposite maxillary processes
- C. Medial and lateral nasal processes
- D. Maxillary and lateral nasal processes
- E. Opposite medial nasal processes



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SOM.MKII.BPM2.1.NB.1.ANAT.0212: Describe the different forms of cleft lip and palate and the developmental processes that fail leading to these defects

A 55-year-old woman comes to the physician because of a 1-month history of persistent dry eyes and dry nasal passages following a skull base surgery. Physical examination shows decreased tear production and dry nasal mucosa, but normal salivation, taste sensation and facial touch sensation. Which of the following structures is most likely affected?

- A. Pterygopalatine ganglion
- B. Chorda tympani
- C. Inferior salivatory nucleus
- D. Superior cervical ganglion
- E. Nerve to stapedius muscle

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A 1-day-old neonate is admitted to the pediatric care unit because of difficulty sucking the breast. Physical examination shows a left posterior cleft palate. Which of the following best describes the embryological development of the secondary palate in this patient

- A. Develops from the second pharyngeal arch
- B. Is formed by the median palatine process
- C. Formed by fusion of the medial nasal processes
- D. Begins as two vertical outgrowths from the developing mandible
- E. Begins as two lateral outgrowths from the developing maxilla

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SOM.MK.I.BPM2.1.NB.1.ANAT.0212 Describe the different forms of cleft lip and palate and the developmental processes that fail leading to these defects

A 6-month-old infant is brought to the pediatrician by her concerned parents. The parents report that their baby has an asymmetrical smile and difficulty feeding. The mother states that during feeding, formula leaks from the right corner of the baby's mouth. Physical examination shows loss of nasolabial fold. These symptoms are caused by a defect to the nerve most likely associated with of the following structures?

- A. 1st pharyngeal arch
- B. 2nd pharyngeal arch
- C. 3rd pharyngeal arch
- D. 4th pharyngeal arch
- E. 5th pharyngeal arch

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SOM.MKII.BPM2.1.NB.1.ANAT.0207: Describe the structures formed by each pharyngeal arch/groove/pouch that contributes to the formation of the face

A 45-year-old man comes to the physician because of a 1-month history of loss of sensation to the lower part of the external ear. He had parotidectomy for removal of a benign pleomorphic adenoma 1-month ago. Physical examination shows decreased sensation over the angle of the mandible and the lower part of his auricle, with intact motor function of the facial muscles. Which of the following nerves is most likely affected in this patient?

- A. Auriculotemporal
- B. Great auricular
- C. Lesser occipital nerve
- D. Cervical dorsal rami 3rd occipital
- E. Buccal branch of Mandibular nerve

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A 47-year-old man comes to the physician because of a 5-day history of swollen upper lip, headaches and generalized facial pain. Physical examination shows a non-fluctuant abscess with diffused erythema on upper lip. Imaging studies show cavernous sinus thrombosis. Which of the following additional symptoms is most likely expected in this patient?

- A. Loss of smell
- B. Loss of sensation to anterior 2/3 of tongue
- C. Loss of sensation to the gingiva of the upper maxillary
- D. Loss of sensation to the gingiva of lower mandible
- E. Dryness of the mouth

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A 45-year-old woman undergoes thyroidectomy for a large multinodular goiter. Postoperatively, she develops difficulty swallowing and a weak cough reflex. Physical examination shows impaired elevation of the pharynx during swallowing. Which of the following muscles is most likely affected?

- A. Cricopharyngeus
- B. Inferior pharyngeal constrictor
- C. Palatopharyngeus
- D. Stylopharyngeus
- E. Superior pharyngeal constrictor

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A 55-year-old woman comes to the physician because of a 2 cm, firm, non-tender nodule in the upper portion of the right lobe of her thyroid gland. She also reports mild dysphagia. Her thyroid function tests are normal and fine needle aspiration of the nodule shows a benign follicular lesion. While resecting the nodule, the resident accidentally traumatises a nerve adjacent to the artery he was trying to ligate. Damage to this nerve typically results in a monotone. The artery that was being ligated is a direct branch of which of the following arteries?

- A. Superior laryngeal
- B. External carotid
- C. Internal carotid
- D. Thyrocervical trunk
- E. Subclavian

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SOM.MKII.BPM2.1.NB.1.ANAT.0074 Describe the branches of the external carotid artery and their areas of distribution

NB IMCQ 4

Histology Questions

1. A 24-year-old man is brought to the emergency department because of a 2-hour history of left eye pain and blurry vision following an accidental blunt head trauma incident. He has a history of Marfan's syndrome. Physical examination shows decreased visual acuity with poor near vision and characteristic musculoskeletal features of Marfan's syndrome. Slit-lamp examination shows dislocation of the lens of the left eye. Dysfunction of the zonular fibers is suspected. These fibers are normally produced and anchored by extensions of the ciliary body. Which of the following is most likely an additional function of these extensions?

- A. Forms the apparatus for the outflow of aqueous humor
- B. Forms part of the blood-aqueous barrier
- C. Regulates the size of the pupil
- D. Participates in light refraction
- E. Forms the site of insertion of the extraocular muscle
- F. Regulates the water content of the corneal stroma

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2. A 45-year-old woman comes to her ophthalmologist for a routine eye exam. She has no complaints of eye discomfort or visual disturbances. During the examination, the ophthalmologist can readily examine the retina through the transparent structure suspended by the zonular fibers. The transparency of this structure and its ability to focus light rays on the retina are due to unique histological features. Which of the following histological features of the eye most likely contributes to the transparency of this structure?

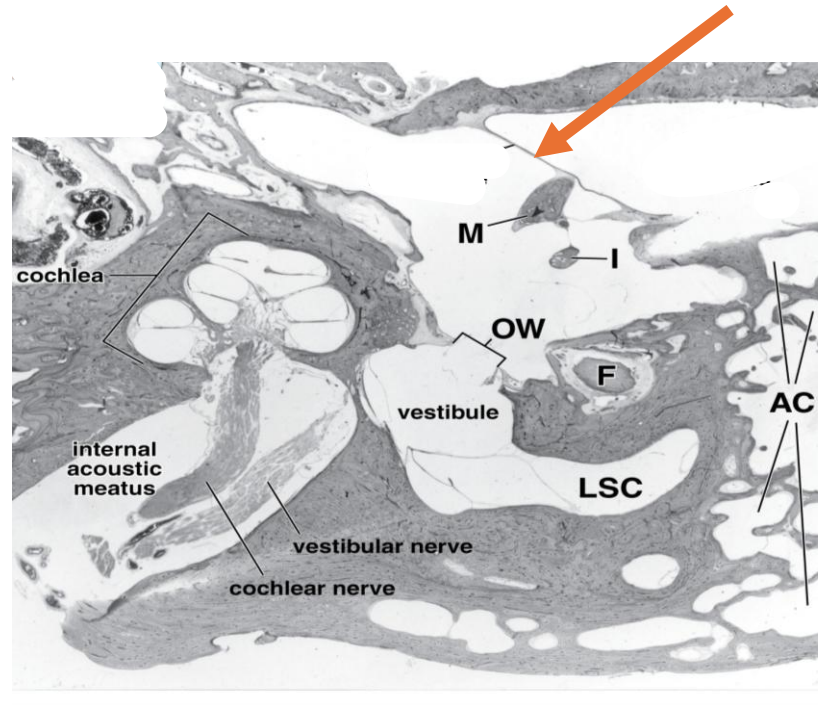
- A. The presence of abundant Na^+ / K^+ ATPase pumps in the lens epithelium
- B. The uniform spacing of collagen fibrils and lamellae
- C. The presence of aqueous humor
- D. The presence of vitreous humor
- E. The organization of lens fibers into tightly packed, anuclear structures
- F. The presence of rods and cones
- G. The presence of the germinal zone at the lens equator

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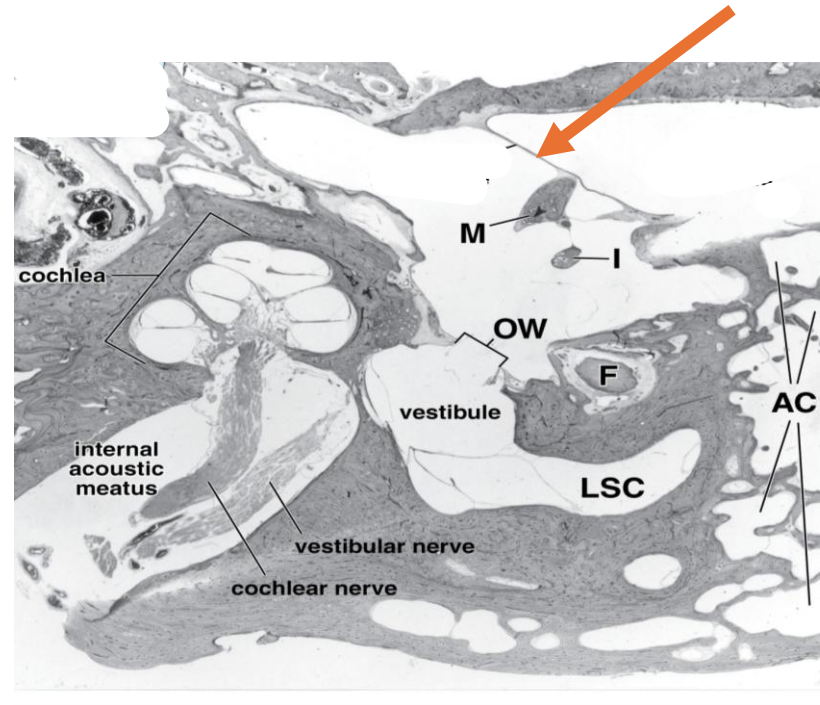
3. A 5-year-old girl is brought to the emergency department because of a 2-day history of right-sided ear pain and fever. She is recovering from an upper respiratory tract infection characterized by rhinorrhea and nasal congestion. Physical examination shows pain exacerbation on traction of the right ear lobe. Otoscopic examination shows hyperemia and bulging of the structure indicated by the arrow in the electron micrograph shown. Which of the following best characterizes the medial surface of this structure?

- A. Radial orientation of collagen fibers
- B. Circular orientation of collagen fibers
- C. Stratified squamous keratinized epithelium
- D. Simple cuboidal epithelium
- E. Presence of eccrine sweat glands
- F. Presence of ceruminous glands



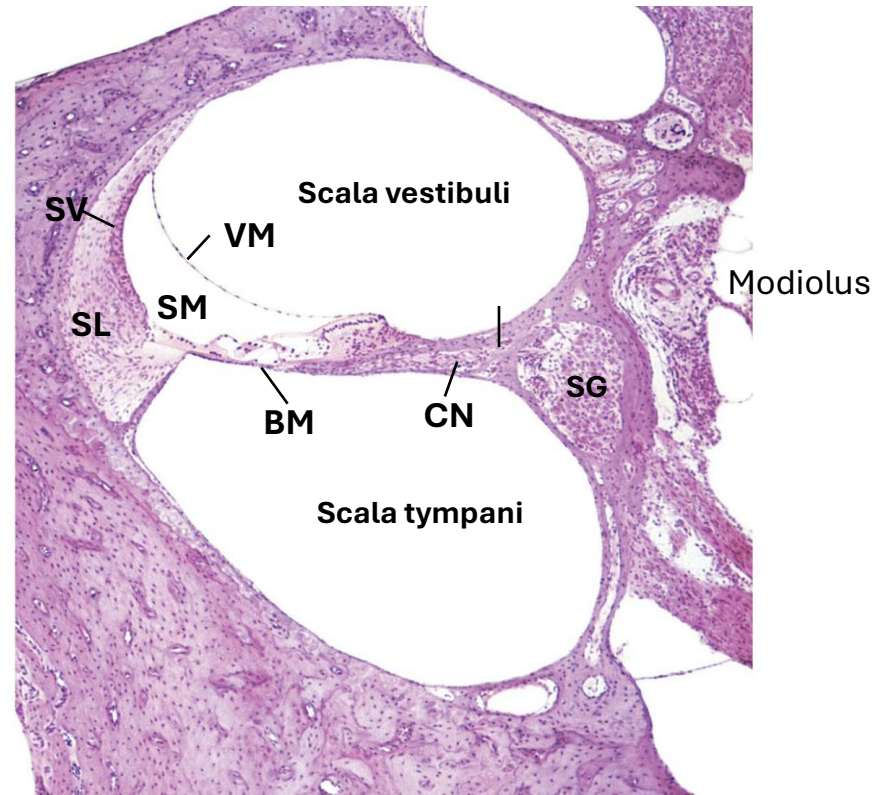
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4. A 53-year-old man comes to the physician because of a 1-week history of bilateral tinnitus (ringing in the ears) and decreased ability to hear. He is a construction worker at a site where he has prolonged exposure to loud noises from heavy duty machinery and does not use protective gear. Physical examination shows no visible abnormalities. Audiometry testing shows bilateral sensorineural hearing loss. The physician suspects damage to the sensory region of the structure shown in the photomicrograph. Which of the following structures is most likely responsible for secreting the fluid that bathes this sensory region?

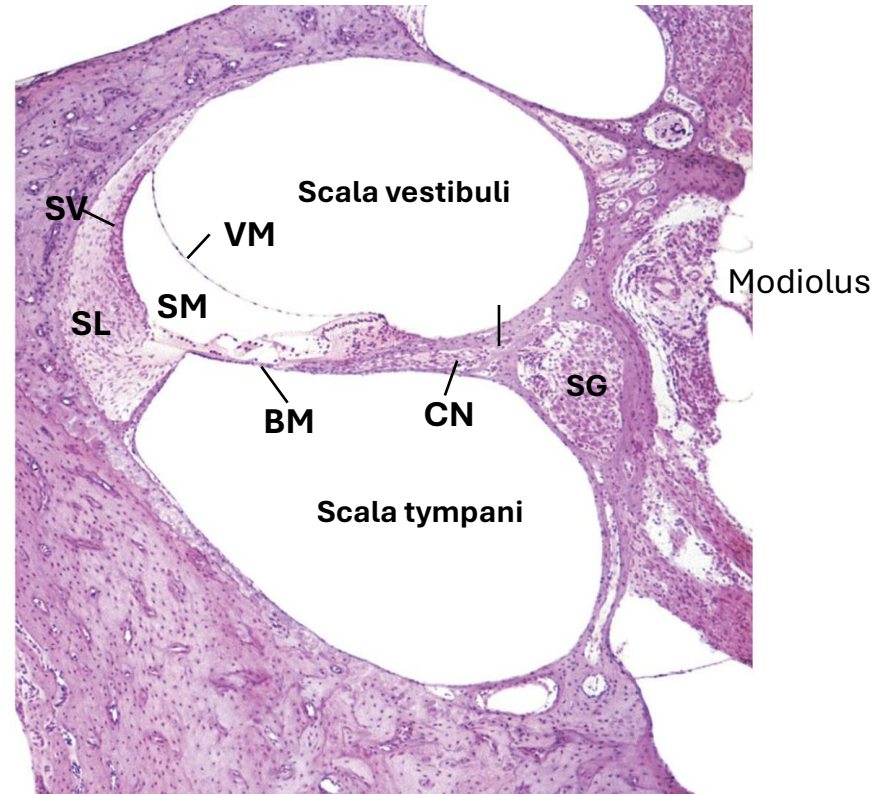
- A. Spiral ganglion
- B. Organ of Corti
- C. Striae vascularis
- D. Periosteum of the cochlea
- E. Tectorial membrane
- F. Basilar membrane
- G. Vestibular membrane



H&E x65

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H&E x65

NB IMCQ 4

Neuroscience

1. A 52-year-old man is brought to the emergency department 30 minutes after being involved in a motor vehicle collision. He says that he has double vision. Neurologic examination shows weakness of the left lateral rectus muscle, consistent with an injury to CN VI. The nucleus of the affected nerve is most likely derived from which of the following embryologically defined plates?

- A. Metencephalic alar
- B. Mesencephalic alar
- C. Myelencephalic alar
- D. Metencephalic basal
- E. Mesencephalic basal
- F. Myelencephalic basal

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2. A 49-year-old man is brought to the physician by his daughter because of a 9-hour history of weakness on the right side of his body. Neuroimaging shows a vascular lesion affecting a structure derived from the ventral mesencephalon. Which of the following structures is most likely lesioned?

- A. Superior colliculus
- B. Precentral gyrus
- C. Pyramid
- D. Crus cerebri
- E. Middle cerebellar peduncle

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3. A 54-year-old woman comes to the physician because of a 6-hour history of numbness of the left side of her face. A CT scan of the head shows a left-sided injury to a cranial nerve. Which of the following embryologically defined structures most likely gives rise to the injured nerve in this patient?

- A. Myelon
- B. Diencephalon
- C. Metencephalon
- D. Telencephalon
- E. Mesencephalon
- F. Myelencephalon

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